

- Response time for specific transactions
- Memory usage
- Disk usage
- CPU utilisation
- System load
- Network interface activity
- Database activities
- Swap space
- Performance of attached disks

Why cloud monitoring is essential

Cloud monitoring is necessary for a number of reasons. The more important ones are listed below.

Security: Security in the cloud is a major roadblock in cloud adoption for business critical applications and in certain industries in which data security is extremely vital.

SLA management: Due to the dynamic nature of the cloud, continuous monitoring on QoS attributes is essential to implement SLAs; and the multifaceted nature of the cloud landscape demands a refined means of managing SLAs.

Capacity planning: Cloud monitoring tools help to identify what resources you are using in the cloud. The performance and capacity requirements can be assessed and, accordingly, resources can be scaled up and down to effectively achieve performance levels and satisfy customers.

Resource management: Resource management is crucial in cloud computing. It maximises resource utilisation and minimises the total cost of both the cloud infrastructure and application hosting.

Trouble shooting: Root cause analysis is a challenging area in cloud computing, considering the involvement of various components and the complex architecture. Cloud monitoring tools can help to diagnose and rectify an issue.

Performance management: Cloud performance management

is the monitoring of resources that manage application performance in cloud environments. It includes supervision of applications to maintain optimal performance and availability.

Billing: Monitoring is a very basic requirement to provide measured services in the cloud environment for CSPs and cloud consumers.

Open source tools for cloud monitoring

Zenoss

Zenoss is an open source platform released under the GNU General Public License (GPL) version 2. It provides an easy-to-use Web UI to monitor performance, events, configuration, and inventory. Zenoss is one of the best for unified monitoring since it is cloud agnostic and is open source. Zenoss provides powerful plug-ins named Zenpacks, which support monitoring on hypervisors (ESX, KVM, Xen and HyperV), private cloud platforms (CloudStack, OpenStack and vCloud/vSphere), and public cloud (AWS).

Hyperic

SpringSource is a division of VMware that has acquired Hyperic, Cloud Foundry, RabbitMQ, and Gemstone. Hyperic can be used for the auto-discovery of all components of virtualised applications. It automatically discovers, manages and monitors IT and network resources on the private cloud (VMware) and public cloud (Amazon Web Services). It is optimised for virtual environments with integration with vCenter and vSphere.

Hyperic provides open source IT resource and network monitoring application software. It auto discovers system resources such as operating systems, hardware, databases, middleware, applications, virtualisation and services. It provides monitoring of network services (SNMP, SMTP, HTTP, and ICMP) and host resources (processor load, disk usage and

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